WORKSHEET

Chapter 3: Optimisation – Cost and revenue

Once you can accept the universe as being something expanding into nothing which is really something …

10	11	12	5	10	1	6

14	7	5	10	13	11	14

9	10	7	8

13	3	12	10	4

10	14

11	12	14	2

Albert Einstein – mathematician

Questions

1 A company makes x video cameras per day at a cost of:

$$C(x) = \frac{x^3}{3} - 100x^2 - 30\,000x + 250.$$

Find the number of video cameras for which production costs are minimised.

2 A company that prints x books per week has costs per week of:

$$C(x) = \frac{x^3}{3} - 300x^2 + 50\,000x + 200.$$

How many books should be produced to minimise costs?

3 A doll manufacturer makes x dolls per day at a cost of:

$$C(x) = 500 - 1800x + 43x^2 - \frac{x^3}{3}.$$

How many dolls should be made to minimise costs?

4 A sports company makes x tennis racquets per day at a cost of:

$$C(x) = x^3 - 381x^2 + 3000x + 100.$$

How many racquets should be made to minimise costs?

A company that manufactures socks receives $5 in revenue per pair. The cost of making x pairs is given by:

$$C(x) = \frac{x^3}{3} - 35x^2 - 795x + 70.$$

5 Write an equation to show the profit made by the company when x pairs are sold.

6 Find the number of pairs that should be manufactured to maximise profit.

A CD manufacturer receives $25 per CD sold. The cost of producing x CDs is given by:

$$C(x) = \frac{x^3}{3} - 226x^2 + 925x + 120.$$

7 Write an equation for the profit made when x CDs are sold.

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8 The sale of how many CDs would maximise profit?

A manufacturer receives \$6000 for each trail bike produced. The cost of producing x bikes per day is:

$C(x) = 50x^3 - 225x^2 + 5400x - 900.$

9 Write an equation for the profit made when x bikes are sold.

10 For what value of x is the profit maximised?

11 What is the maximum profit?

The revenue received from the sale of x TVs is: $R(x) = 300x$. The production cost of x TVs is:

$$C(x) = 3000 - 720x + 32x^2 - \frac{x^3}{3}.$$

12 Write an equation for the profit made when x TVs are sold.

13 For what value of x is the profit maximised?

14 What is the maximum profit?

Solutions

A $P(x) = \frac{x^3}{3} - 32x^2 + 1020x - 3000$

D 250

E \$3700

G 80

H 450

I 4

L 36

N 300

P 30

R $P(x) = -70 + 800x + 35x^2 - \frac{x^3}{3}$

S \$7800

T $P(x) = -120 - 900x + 226x^2 - \frac{x^3}{3}$

W $P(x) = 900 + 600x + 225x^2 - 50x^3$

Y 500